

Development of Laser Bending Strategy with IoT-Based Sensors and Control

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Figure 1: Trident CNC based LASER machine with Sinumerik 828D controller

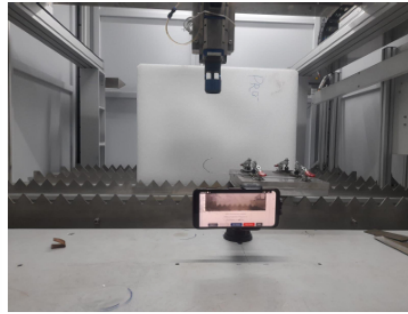


Figure 2: Apparatus setup image

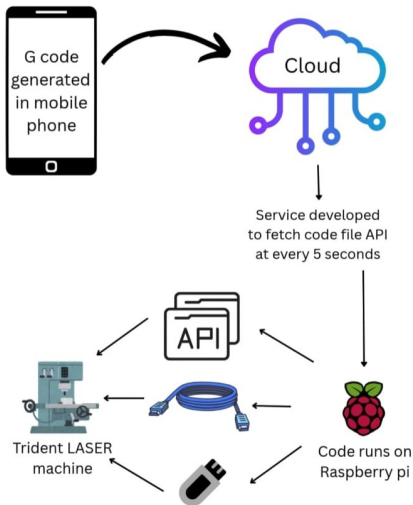


Figure 3: Overall system pipeline illustrating the flow from data acquisition to adaptive control, adapted from project presentation overview.

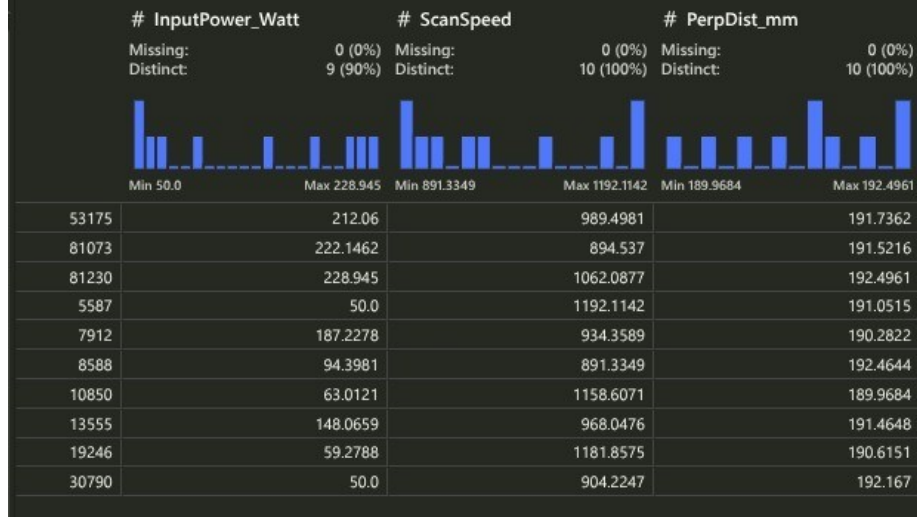


Figure 4: FCAL analysis over user provided CAD model and the profile generated using ML predicted parameters.

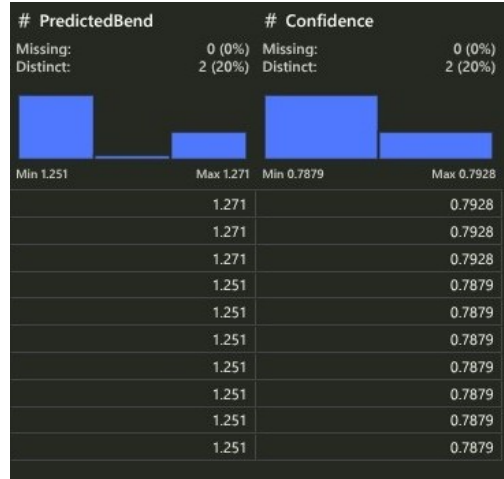


Figure 5: FCAL analysis continued

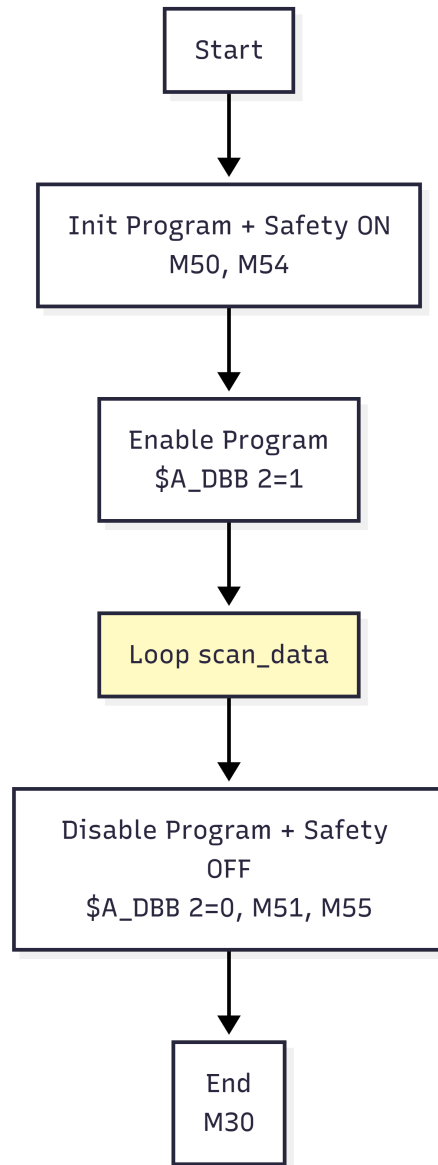


Figure 6: CNC Code generation pipeline

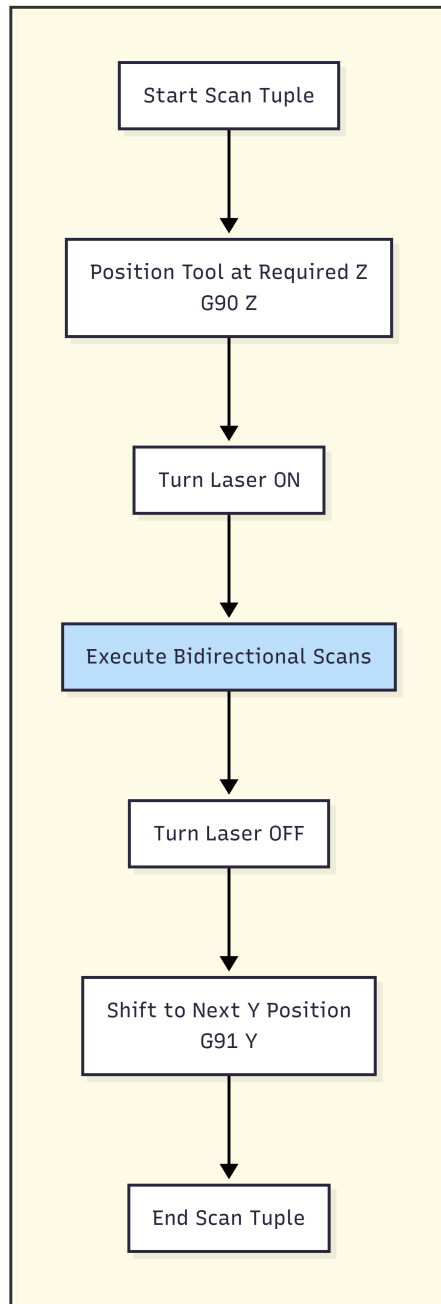


Figure 7: Loop scan data

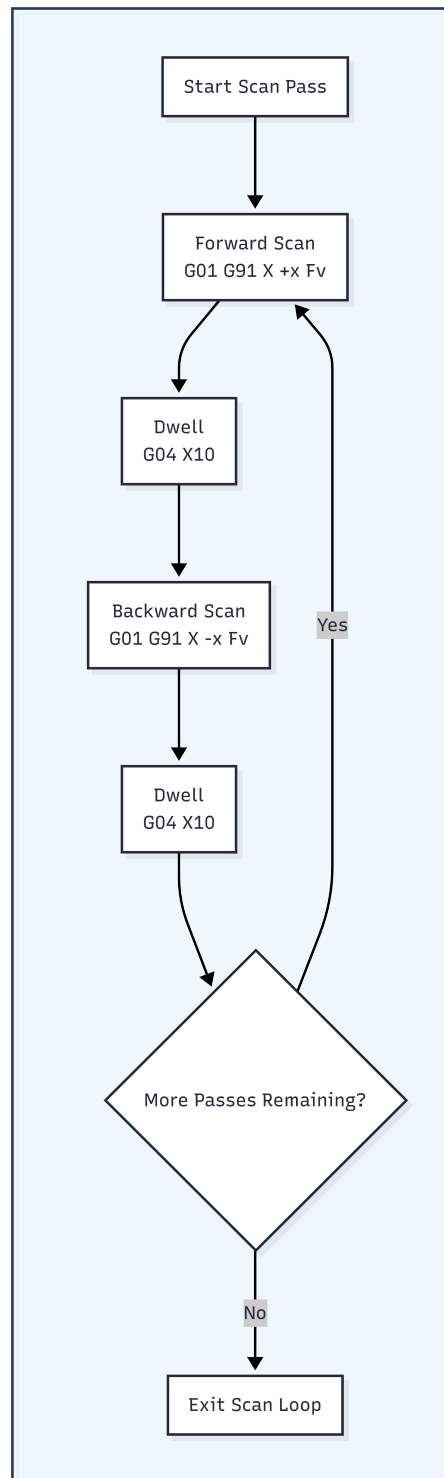


Figure 8: bidirectional scans

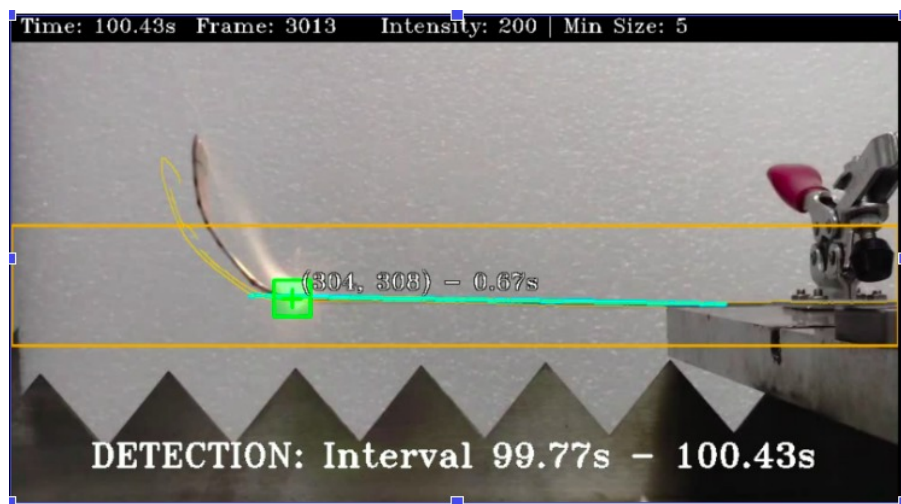


Figure 9: Sample experimental dataset



Figure 10: Model Generated Parameters, used in evaluation scores.



Figure 11: Evaluation image scan 1



Figure 12: Evaluation image scan 2

timeline of the experiment	scan ID	thickness (in mm)	width (in mm)	actual_lase_power (watt)
february 11, 2025 starts	1A	0.8	25	340
	2A	0.8	25	340
	3A	0.8	25	340
	4A	0.8	25	340
february 11, 2025 ends	5A	0.8	25	340
february 15, 2025 starts	6A	0.8	25	340
	7A	0.8	25	340
	8A	0.8	25	340
	9A	0.8	25	340
february 15, 2025 ends	10A	0.8	25	340
march 2, 2025 starts	1B	0.8	25	110
	2B	0.8	25	120
	3B	0.8	25	160
	4B	0.8	25	190
march 2, 2025 ends	5B	0.8	25	230

Figure 13: Sample experimental dataset

Desired angle value	InputPower_Watt	ScanSpeed	PerpDist_mm	RefDist_mm	Curr_Iter	PredictedBend	Confidence	s.no
2.3	292.162734	2000	180.383348	15	7	2.300029	0.446497	
1.5	500	1168.882524	191.551782	6.353097	6	1.500212	0.519694	2
2.75	346.643566	700	184.556852	5	1	2.749895	0.424975	1
2	500	1990.016783	210	9.109154	4	2.000102	0.413649	3
3.56	367.761907	2000	174.331867	9.803671	1	3.574895	0.399155	4
3.855	500	2000	165.327816	13.794895	1	3.855711	0.522298	5

Figure 14: Model Generated Parameters, used in evaluation scores.

input power to machine	scan speed	number of scans	perpendicular distance of scanning	distance from reference point
500	1500	6	190	0
500	1500	6	195	+5
500	1500	6	200	+5
500	2000	6	160	+5
500	2000	6	170	+5
500	2000	6	180	+5
500	2000	7	188	+5
500	2000	6	190	+5
500	2000	6	195	+5
500	2000	6	200	+5
265	10000	8	175	0
280	10000	8	175	+5
320	10000	8	190	+5
360	10000	8	210	+5
395	10000	3	180	+5

Figure 15: development of image mask using OpenCV for scanning detection